

Limit ve Süreklilik (Soru Çözümü)

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Limit ve
Süreklilik...

Soru: $\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\sqrt{x + \sqrt{x + \sqrt{x}}}} = ? \frac{\infty}{\infty}$

$$x + x \sqrt{\frac{1}{x}}$$

$$x \left(1 + \frac{1}{\sqrt{x}} \right)$$

$$\sqrt{x}$$

$$\sqrt{x + \sqrt{x}}$$

$$x \left(1 + \frac{1}{\sqrt{x}} \right)$$

$$\sqrt{x}$$

$\frac{0}{0}, \frac{\infty}{\infty}, \infty - \infty, 0 \cdot \infty, \frac{\infty}{1}, \frac{\infty}{\infty}, \frac{0}{0}$
Belirsiz

$$\lim_{x \rightarrow \infty} \frac{\sqrt{x}}{\sqrt{x}} = 1$$

Soru: $\lim_{x \rightarrow 1} \frac{x^3 - 3x + 2}{x^4 - 4x + 3} = ? \left(\frac{0}{0} \right)$ belirsizliği var.

$$\begin{array}{r} x^3 - 3x + 2 \quad | \quad x-1 \\ x^3 - x^2 \\ \hline x^2 - 3x + 2 \\ x^2 - x \\ \hline -2x + 2 \\ -2x + 2 \\ \hline 0 \end{array}$$

$$\begin{array}{r} x^4 - 4x + 3 \quad | \quad x-1 \\ x^4 - x^3 \\ \hline x^3 - 4x + 3 \\ x^3 - x^2 \\ \hline x^2 - 4x + 3 \\ x^2 - x \\ \hline -3x + 3 \\ -3x + 3 \\ \hline 0 \end{array}$$

$$\lim_{x \rightarrow 1} \frac{(x-1)(x^2+x-2)}{(x-1)(x^3+x^2+x-3)} = \frac{0}{0}$$

$$\begin{array}{r} x^3 + x^2 + x - 3 \quad | \quad x-1 \\ x^3 - x^2 \\ \hline 2x^2 + x - 3 \\ 2x^2 - 2x \\ \hline 3x - 3 \\ 3x - 3 \\ \hline 0 \end{array}$$

$$\lim_{x \rightarrow 1} \frac{(x+2)(x-1)}{(x^2+2x+3)(x-1)} = \frac{3}{6} = \frac{1}{2} //$$

Soru: $\lim_{x \rightarrow \infty} x(\sqrt{x^2+1}-x) = ?$ $\infty(\infty-\infty)$ belirsizlik

$$\begin{aligned} &= \lim_{x \rightarrow \infty} \frac{x(\sqrt{x^2+1}-x)(\sqrt{x^2+1}+x)}{\sqrt{x^2+1}+x} = \lim_{x \rightarrow \infty} \frac{x(\cancel{x^2+1}-\cancel{x^2})}{\sqrt{x^2+1}+x} \\ &= \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2+1}+x} \quad \left(\frac{\infty}{\infty}\right) \\ &\quad \swarrow x^2\left(1+\frac{1}{x^2}\right) \\ &= \lim_{x \rightarrow \infty} \frac{x}{x\sqrt{1+\frac{1}{x^2}}+x} \\ &= \lim_{x \rightarrow \infty} \frac{\cancel{x}}{\cancel{x}\left(\sqrt{1+\frac{1}{x^2}}+1\right)} \\ &= \frac{1}{2}, \end{aligned}$$

Soru: $\lim_{x \rightarrow 64} \frac{\sqrt{x} - 8}{\sqrt[3]{x} - 4} = ?$ $\frac{0}{0}$ belirsizlik!

I.yol $x = t^6$

$x \rightarrow 64, t \rightarrow 2$

$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

$$\lim_{t \rightarrow 2} \frac{\sqrt{t^6} - 8}{\sqrt[3]{t^6} - 4} = \lim_{t \rightarrow 2} \frac{t^3 - 8}{t^2 - 4} \quad \left(\frac{0}{0} \right)$$

$$= \lim_{t \rightarrow 2} \frac{(t-2)(t^2 + 2t + 4)}{(t-2)(t+2)}$$

$$= \frac{12}{4} = 3 \quad \checkmark$$

$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$
 $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$?

II.yol : $\lim_{x \rightarrow 64} \frac{\sqrt{x} - 8}{\sqrt[3]{x} - 4} \cdot \frac{\sqrt{x} + 8}{\sqrt{x} + 8} \cdot \frac{\sqrt[3]{x^2 + 4\sqrt{x} + 16}}{\sqrt[3]{x^2 + 4\sqrt{x} + 16}}$

$$= \lim_{x \rightarrow 64} \frac{(x-64)(\sqrt[3]{x^2 + 4\sqrt{x} + 16})}{(x-64)(\sqrt{x} + 8)} = \frac{16 + 16 + 16}{16} = 3$$

Soru: $\lim_{x \rightarrow -2^+} \frac{\tan(\pi x)}{x+2} = ?$ $\frac{\tan(-2\pi)}{0} = \frac{0}{0}$ belirsizlik

$x = -2 + h$
 $x \rightarrow -2^+, h \rightarrow 0$

$$\begin{aligned}
 &= \lim_{x \rightarrow -2^+} \frac{\sin \pi x}{\cos \pi x \cdot (x+2)} = \lim_{x \rightarrow -2^+} \frac{\sin \pi x}{x+2} \cdot \lim_{x \rightarrow -2^+} \frac{1}{\cos \pi x} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(\pi(-2+h))}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin(-2\pi + \pi h)}{h} \\
 &= \lim_{h \rightarrow 0} \frac{\sin \pi h}{h} \\
 &= \pi
 \end{aligned}$$

$\sin(-2\pi)\cos \pi h + \cos(-2\pi)\sin \pi h$

Dönüşüm Formülleri

$$\sin x + \sin y = 2 \cdot \sin\left(\frac{x+y}{2}\right) \cdot \cos\left(\frac{x-y}{2}\right)$$

$$\sin x - \sin y = 2 \cdot \cos\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{x-y}{2}\right)$$

$$\cos x + \cos y = 2 \cdot \cos\left(\frac{x+y}{2}\right) \cdot \cos\left(\frac{x-y}{2}\right)$$

$$* \cos x - \cos y = -2 \cdot \sin\left(\frac{x+y}{2}\right) \cdot \sin\left(\frac{x-y}{2}\right)$$

$$\begin{aligned}
 \cos(a+b) &= \cos a \cdot \cos b - \sin a \cdot \sin b \\
 \cos(a-b) &= \cos a \cdot \cos b + \sin a \cdot \sin b
 \end{aligned}$$

$$\cos(a+b) - \cos(a-b) = -2 \sin a \cdot \sin b$$

$$\tan x + \tan y = \frac{\sin(x+y)}{\cos x \cdot \cos y}$$

$$\tan x - \tan y = \frac{\sin(x-y)}{\cos x \cdot \cos y}$$

$$\cot x + \cot y = \frac{\sin(x+y)}{\sin x \cdot \sin y}$$

$$\cot x - \cot y = \frac{\sin(y-x)}{\sin x \cdot \sin y}$$

$$\begin{aligned}
 a+b &= x \Rightarrow a = \frac{x+y}{2}, b = \frac{x-y}{2} \\
 a-b &= y
 \end{aligned}$$

$$\cos x - \cos y = -2 \sin\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right)$$

$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

Soru: $\lim_{x \rightarrow a} \frac{\cos x - \cos a}{x - a} = ?$ $\frac{\cos a - \cos a}{a - a} = \frac{0}{0}$ belirsizliği

$$= \lim_{x \rightarrow a} \frac{-2 \sin\left(\frac{x+a}{2}\right) \sin\left(\frac{x-a}{2}\right)}{x-a}$$

$$= - \lim_{x \rightarrow a} \sin\left(\frac{x+a}{2}\right) \cdot \lim_{x \rightarrow a} \frac{\sin\left(\frac{x-a}{2}\right)}{\frac{x-a}{2}}$$

$$\frac{x-a}{2} = t \quad \lim_{t \rightarrow 0} \frac{\sin t}{t} = 1$$

$$x \rightarrow a, \quad t \rightarrow 0$$

$\sin a$

$$= \underline{\underline{-\sin a}}$$

Soru: $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x^3}\right) = ?$ 0. ✓

?

$$-1 \leq \cos\left(\frac{1}{x^3}\right) \leq 1$$

$$-x^2 \leq x^2 \cos\left(\frac{1}{x^3}\right) \leq x^2$$

$\lim_{x \rightarrow 0} (-x^2) = 0$
 $\lim_{x \rightarrow 0} x^2 = 0$
 $\lim_{x \rightarrow 0} x^2 \cos\left(\frac{1}{x^3}\right) = 0$

Soru: $\lim_{x \rightarrow \frac{\pi}{8}} \frac{8x - \pi}{\sin 2x - \cos 2x} = ?$ $\frac{\frac{\pi}{4} - \frac{\pi}{4}}{\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}} = \frac{0}{0} \checkmark$

$\cos\left(\frac{\pi}{2} - 2x\right)$

$$\begin{aligned}\cos\left(\frac{\pi}{2} - 2x\right) - \cos 2x &= -2 \sin\left(\frac{\frac{\pi}{2} - 2x + 2x}{2}\right) \cdot \sin\left(\frac{\frac{\pi}{2} - 2x - 2x}{2}\right) \\ &= -2 \sin\left(\frac{\pi}{4}\right) \cdot \sin\left(\frac{\pi}{4} - 2x\right) \\ &= -\sqrt{2} \sin\left(\frac{\pi}{4} - 2x\right)\end{aligned}$$

$\lim_{x \rightarrow \frac{\pi}{8}} \frac{8x - \pi}{-\sqrt{2} \sin\left(\frac{\pi}{4} - 2x\right)} = \frac{0}{0} \checkmark$

$\lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$

$\lim_{\theta \rightarrow 0} \frac{\theta}{\sin \theta} = 1$

$x - \frac{\pi}{8} = t \Rightarrow 8x - \pi = 8t$
 $\frac{\pi}{4} - 2x = -2t$
 $x \rightarrow \frac{\pi}{8}, t \rightarrow 0$

$$= -\frac{1}{\sqrt{2}} \lim_{t \rightarrow 0} \frac{8t}{\sin(-2t)} = -\frac{4}{\sqrt{2}} \lim_{t \rightarrow 0} \frac{(-2t)}{\sin(-2t)} = \frac{4}{\sqrt{2}} = \underline{\underline{2\sqrt{2}}}$$

1

Soru: $\lim_{x \rightarrow \frac{\pi}{3}^+} \frac{1-2\cos x}{\pi-3x} = ?$ $\frac{1-2\cos\frac{\pi}{3}}{\pi-3(\frac{\pi}{3})} = \frac{0}{0}$, belirsizliği

$$x = \frac{\pi}{3} + h$$

$$x \rightarrow \frac{\pi}{3}^+, h \rightarrow 0$$

$$= \lim_{h \rightarrow 0} \frac{1-2\cos\left(\frac{\pi}{3}+h\right)}{\pi-3\left(\frac{\pi}{3}+h\right)} = \lim_{h \rightarrow 0} \frac{1-2\left(\cos\frac{\pi}{3} \cdot \cos h - \sin\frac{\pi}{3} \cdot \sin h\right)}{\pi-\pi-3h}$$

$$= \lim_{h \rightarrow 0} \frac{1-\cos h + \sqrt{3}\sin h}{-3h} \quad \left(\frac{0}{0}\right)$$

$$= -\frac{1}{3} \lim_{h \rightarrow 0} \frac{1-\cos h}{h} - \frac{\sqrt{3}}{3} \lim_{h \rightarrow 0} \frac{\sin h}{h} = -\frac{\sqrt{3}}{3}$$

$$\cos h = 1 - 2\sin^2 \frac{h}{2}$$

$$\lim_{h \rightarrow 0} \frac{1 - (1 - 2\sin^2 \frac{h}{2})}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2\sin^2 \frac{h}{2}}{h} = \lim_{h \rightarrow 0} \frac{\sin^2 \frac{h}{2}}{\frac{h}{2}} = \lim_{h \rightarrow 0} \underbrace{\frac{\sin \frac{h}{2}}{\frac{h}{2}}}_1 \cdot \underbrace{\lim_{h \rightarrow 0} \sin \frac{h}{2}}_0 = 0$$

BİR BİLGİ

$$\lim_{x \rightarrow \infty} (1 + \underbrace{u(x)}_0)^{\underbrace{v(x)}_{\infty}}$$

$$\left. \begin{array}{l} \lim_{x \rightarrow \infty} u(x) = 0 \quad \checkmark \\ \lim_{x \rightarrow \infty} v(x) = \infty \quad \checkmark \\ \lim_{x \rightarrow \infty} u(x)v(x) = k \quad \checkmark \end{array} \right\}$$

$$\Rightarrow \lim_{x \rightarrow \infty} (1 + u(x))^{v(x)} = e^k$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e \quad \checkmark \checkmark$$

$$x = \frac{1}{t} \Rightarrow x \rightarrow \infty, t \rightarrow 0 \Rightarrow$$

$$\lim_{t \rightarrow 0} (1 + t)^{\frac{1}{t}} = e$$

$$\left. \begin{array}{l} \lim_{x \rightarrow \infty} \frac{1}{x} = 0 \\ \lim_{x \rightarrow \infty} x = \infty \end{array} \right\} \lim_{x \rightarrow \infty} \frac{1}{x} \cdot x = 1$$

$$\frac{1}{x} = t \quad \begin{array}{c} \infty \\ 1 \end{array}$$

Soru: $\lim_{x \rightarrow \infty} \left(\frac{3x+5}{3x-3} \right)^{4x-3} = ?$

$$\frac{3x+5}{3x-3} = 1 + \frac{8}{3x-3} \quad \begin{array}{r} 3x+5 \\ - 3x-3 \\ \hline 8 \end{array} \quad \frac{3x+5}{3x-3} = 1 + \frac{8}{3x-3}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{8}{3x-3} \right)^{4x-3}$$

$\underbrace{\quad}_{u(x)}$

$$\lim_{x \rightarrow \infty} \frac{8}{3x-3} = 0 \quad \checkmark$$

$$\lim_{x \rightarrow \infty} 4x-3 = \infty \quad \checkmark$$

$$\lim_{x \rightarrow \infty} \left(\frac{8}{3x-3} \right) \cdot (4x-3) = \lim_{x \rightarrow \infty} \frac{32x-24}{3x-3} = \frac{32}{3}$$

$$\lim_{x \rightarrow \infty} \left(1 + \frac{8}{3x-3} \right)^{4x-3} = e^{\frac{32}{3}} \quad \checkmark$$

Soru: $\lim_{x \rightarrow 0} \frac{\ln(x+1)}{2^x - 1} = ?$ $\left(\frac{0}{0}\right)$

$$= \lim_{x \rightarrow 0} \frac{\ln(x+1)}{2^x - 1} \cdot \frac{x}{x} = \lim_{x \rightarrow 0} \frac{\ln(x+1)}{x} \cdot \lim_{x \rightarrow 0} \frac{x}{2^x - 1} = \frac{1}{\ln 2}$$

$$\lim_{x \rightarrow 0} \frac{\ln(x+1)}{x} = \lim_{x \rightarrow 0} \left(\ln(x+1)^{\frac{1}{x}} \right) = \ln \left(\lim_{x \rightarrow 0} (x+1)^{\frac{1}{x}} \right) = \ln e = 1$$

$$\lim_{x \rightarrow 0} \frac{x}{2^x - 1} \Rightarrow 2^x - 1 = t, \quad x \rightarrow 0, \quad t \rightarrow 0 \quad = \lim_{t \rightarrow 0} \frac{1}{\ln 2} \cdot \frac{\ln(t+1)}{t} \checkmark$$

$$\ln(2^x) = \ln(t+1)$$

$$x \cdot \ln 2 = \ln(t+1)$$

$$x = \frac{1}{\ln 2} \ln(t+1)$$

$$\ln a^b = b \ln a$$

$$\ln(t+1)^{\frac{1}{t}} = \frac{1}{t} \ln(t+1)$$

$$= \frac{1}{\ln 2} \lim_{t \rightarrow 0} \left(\ln(t+1)^{\frac{1}{t}} \right)$$

$$= \frac{1}{\ln 2} \ln \left(\lim_{t \rightarrow 0} (t+1)^{\frac{1}{t}} \right)$$

$$= \frac{1}{\ln 2}$$

Soru: $\lim_{x \rightarrow e} \frac{e-x}{1-\ln x} = ? \quad \frac{0}{0}$

$$= \lim_{x \rightarrow e} \frac{e-x}{\ln e - \ln x} = \lim_{x \rightarrow e} \frac{x-e}{\ln x - \ln e} = \lim_{x \rightarrow e} \frac{x-e}{\ln\left(\frac{x}{e}\right)}$$

$\frac{x}{e} = t+1, \quad x = e(t+1)$
 $x \rightarrow e, \quad t \rightarrow 0$

$$= \lim_{t \rightarrow 0} \frac{e \cdot t + \cancel{e} - \cancel{e}}{\ln(t+1)}$$

$$= e \cdot \lim_{t \rightarrow 0} \frac{t}{\ln(t+1)}$$

$$= e \cdot \lim_{t \rightarrow 0} \left(\frac{\ln(t+1)}{t} \right)^{-1}$$

$$= e \cdot \lim_{t \rightarrow 0} \left(e^{(t+1) \frac{1}{t}} \right)^{-1}$$

$$= e \left[\ln \left(\underbrace{\lim_{t \rightarrow 0} (t+1)^{\frac{1}{t}}}_e \right) \right]^{-1}$$

$$\underbrace{\hspace{10em}}_1$$

$$= \underline{\underline{e}}$$

Soru: $\lim_{x \rightarrow 0} \frac{\cos^3 3x - \cos^3 2x}{x^2} = ?$ $\frac{0}{0}$

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2)$$

$$= \lim_{x \rightarrow 0} \frac{(\cos 3x - \cos 2x) (\cos^2 3x + \cos 3x \cos 2x + \cos^2 2x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\cos 3x - \cos 2x}{x^2} \cdot \lim_{x \rightarrow 0} (\overbrace{\cos^2 3x}^1 + \overbrace{\cos 3x \cos 2x}^1 + \overbrace{\cos^2 2x}^1)$$

$$= 3 \lim_{x \rightarrow 0} \frac{\cos 3x - \cos 2x}{x^2} \quad \left(\frac{0}{0} \right)$$

$$\cos a - \cos b = -2 \sin\left(\frac{a+b}{2}\right) \sin\left(\frac{a-b}{2}\right)$$

$$= 3 \cdot \lim_{x \rightarrow 0} \frac{-2 \cdot \sin\left(\frac{6x}{2}\right) \cdot \sin\left(\frac{x}{2}\right)}{x^2}$$

$$= -6 \lim_{x \rightarrow 0} \underbrace{\frac{\sin\left(\frac{5x}{2}\right)}{x}}_{\frac{5}{2}} \cdot \lim_{x \rightarrow 0} \underbrace{\frac{\sin\left(\frac{x}{2}\right)}{x}}_{\frac{1}{2}} = -6 \cdot \frac{5}{2} \cdot \frac{1}{2} = -\frac{15}{2} \checkmark$$

Soru: $\lim_{x \rightarrow 0} \frac{\sin \alpha x - \tan \alpha x}{x^3} = ?$ $\frac{0}{0}$

$$= \lim_{x \rightarrow 0} \frac{\sin \alpha x - \frac{\sin \alpha x}{\cos \alpha x}}{x^3} = \lim_{x \rightarrow 0} \frac{\sin \alpha x \left(1 - \frac{1}{\cos \alpha x}\right)}{x^3} = \lim_{x \rightarrow 0} \frac{\sin \alpha x \cdot (\cos \alpha x - 1)}{\cos \alpha x \cdot x^3}$$

$$= \underbrace{\lim_{x \rightarrow 0} \frac{\sin \alpha x}{x}}_{\alpha} \cdot \lim_{x \rightarrow 0} \frac{\cos \alpha x - 1}{x^2} \cdot \underbrace{\lim_{x \rightarrow 0} \frac{1}{\cos \alpha x}}_1$$

$$= \alpha \lim_{x \rightarrow 0} \frac{\cos \alpha x - 1}{x^2} \quad \left(\frac{0}{0}\right)$$

$$= \alpha \lim_{x \rightarrow 0} \frac{\cancel{1} - 2\sin^2 \frac{\alpha x}{2} - \cancel{1}}{x^2}$$

$$= -2\alpha \cdot \underbrace{\lim_{x \rightarrow 0} \frac{\sin \frac{\alpha x}{2}}{x}}_{\alpha/2} \cdot \underbrace{\lim_{x \rightarrow 0} \frac{\sin \frac{\alpha x}{2}}{x}}_{\alpha/2} = -2\alpha \cdot \frac{\alpha}{2} \cdot \frac{\alpha}{2} = -\frac{\alpha^3}{2} //$$

$\cos \alpha x = 1 - 2\sin^2 \frac{\alpha x}{2}$

Soru: $\lim_{x \rightarrow \infty} \frac{3x}{\lfloor x \rfloor} = ?$ $\frac{\infty}{\infty}$

$$\frac{\lfloor x \rfloor}{\lfloor x \rfloor} < \frac{x}{\lfloor x \rfloor} < \frac{\lfloor x \rfloor + 1}{\lfloor x \rfloor}$$

$$1 < \frac{x}{\lfloor x \rfloor} < 1 + \frac{1}{\lfloor x \rfloor}$$

$$3 < \frac{3x}{\lfloor x \rfloor} < 3 + \frac{3}{\lfloor x \rfloor}$$

$\lim_{x \rightarrow \infty} 3 = 3$
 $\lim_{x \rightarrow \infty} \frac{3x}{\lfloor x \rfloor} = 3$ ✓

$\lim_{x \rightarrow \infty} 3 + \frac{3}{\lfloor x \rfloor} = 3$

Soru: $\lim_{x \rightarrow \frac{\pi}{4}} [\tan 2x (\tan x - 1)] = ?$ $(\infty \cdot 0)$ belirsizliği

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin 2x}{\cos 2x} \cdot \left(\frac{\sin x}{\cos x} - 1 \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x} \cdot \left(\frac{\sin x - \cos x}{\cos x} \right)$$

$$= \lim_{x \rightarrow \frac{\pi}{4}} \frac{2 \cancel{\sin x} \cancel{\cos x}}{(\cancel{\cos x} - \cancel{\sin x})(\cos x + \sin x)} \cdot \frac{\cancel{\sin x} - \cancel{\cos x}}{\cancel{\cos x}} = - \frac{2 \frac{\sqrt{2}}{2}}{\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}} = - \frac{\sqrt{2}}{\sqrt{2}} = -1$$

Soru: $f: \mathbb{R} \rightarrow \mathbb{R}$ $f(x) = \begin{cases} \frac{\sin 2x}{x} & \underline{x < 0} \\ m & \underline{x = 0} \\ \frac{3x-n}{x+3} & \underline{x > 0} \end{cases}$ fonksiyonunun sürekli olması için m ve n ne

olmalıdır.?

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$\lim_{x \rightarrow 0^-} \frac{\sin 2x}{x} = \lim_{x \rightarrow 0^+} \frac{3x-n}{x+3} = m$$

$$2 = -\frac{n}{3} = m \quad \Rightarrow \quad \underline{m = 2}, \quad \underline{n = -6}$$

Soru: $f(x) = \frac{3x^2 - 4x - 1}{x^2 - (m+1)x + 1}$ fonksiyonu hangi m değerleri için süreklidir?

??

$\checkmark \Delta > 0, \checkmark \Delta = 0, \Delta < 0$ ***

$$b^2 - 4ac < 0$$

✓

$$(-(m+1))^2 - 4 \cdot 1 \cdot 1 < 0$$

$$m^2 + 2m + 1 - 4 < 0$$

$$m^2 + 2m - 3 < 0$$

$$(m+3)(m-1) < 0$$

$$m_1 = -3, m_2 = 1$$

$$\begin{array}{c|ccc} m & -3 & & 1 \\ \hline & + & - & + \end{array}$$

$m \in (-3, 1)$ fonksiyon süreklidir //

Soru: Aşağıdaki fonksiyonların süreksizlik çeşitlerini bulunuz?

(a) • $f(x) = \frac{x^2}{x^2 - 3x + 2} \Rightarrow x^2 - 3x + 2 = 0$ $\lim_{x \rightarrow 1} \frac{x^2}{x^2 - 3x + 2} = \frac{1}{0} = \infty$
 $(x-2)(x-1) = 0$
 $x=2, x=1$ $\lim_{x \rightarrow 2} \frac{x^2}{x^2 - 3x + 2} = \frac{4}{0} = \infty$
 • $g(x) = \frac{x^3 - 1}{x^2 - 1}$
 • $h(x) = \frac{|x|}{x}$
 $x=1, x=2$, Sonuç sınırsızlığı

(b) $x^2 - 1 = 0$
 $x = \pm 1$, $\lim_{x \rightarrow a} f(x) \neq f(a)$
 $\lim_{x \rightarrow 1} \frac{x^3 - 1}{x^2 - 1} = \left(\frac{0}{0}\right)$, $\lim_{x \rightarrow 1} \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{3}{2}$, kaldırılabilir süreksizlik
 $\lim_{x \rightarrow -1} \frac{x^3 - 1}{x^2 - 1} = \lim_{x \rightarrow -1} \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)} = \frac{1}{0} = \infty$, $\Rightarrow x = -1$, sonuç sınırsızlığı

(c) $h(x) = \frac{|x|}{x}$, $x=0$, kritik nokta
 $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = \lim_{x \rightarrow 0^+} \frac{x}{x} = 1$
 $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = \lim_{x \rightarrow 0^-} \frac{-x}{x} = -1$
 $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$
 $x=0$, sıçrama süreksizliği